Marketing Analytics

Experiments and Incremental Value

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Session 09



An Email-Campaign Decision

- Eligible customers receive either an email or no email.
- Measure purchases over a fixed response window.
- The decision concerns contribution per assigned customer.

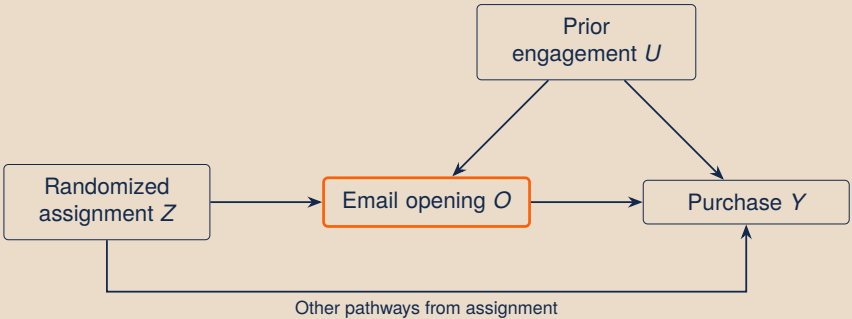
Potential Outcomes

For eligible customer *i*, define outcomes with and without assignment.

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Each customer has one observed outcome under the realized assignment.

Selection Creates an Alternative Explanation



Opening depends on engagement and assignment.
Keep all assigned customers in the analysis.

Randomized assignment is distinct from engagement-driven email opening.

Email Assignment and Purchase

Assigned Group	Customers	Purchases	Rate
Email	1,000	130	13%
Control	1,000	100	10%

Synthetic experiment. Each customer has one binary outcome.

An Assignment-Effect Estimate

Compare purchase rates across all assigned customers.

$$\widehat{\tau} = \frac{130}{1000} - \frac{100}{1000} = 0.03$$

The estimated effect is three additional purchases per 100 assigned customers.

Sampling Uncertainty

Use an independent-group approximation for the synthetic binary outcomes.

$$\widehat{\mathrm{SE}} = \sqrt{\frac{.13(.87)}{1000} + \frac{.10(.90)}{1000}} \approx .0143$$

An approximate 95% interval is 0.21 to 5.79 percentage points.

Contribution and Decision Uncertainty

- An extra purchase contributes 40 dollars.
- Delivery costs 0.50 dollars per assigned customer.
- Estimated incremental contribution is 0.70 dollars per customer.

The Experiment Calculation

Reproduce the effect and its contribution implication.

```
from math import sqrt
p1, p0, n = 0.13, 0.10, 1000
effect = p1 - p0
se = sqrt(p1*(1-p1)/n + p0*(1-p0)/n)
interval = (effect - 1.96*se, effect + 1.96*se)
profit = 40 * effect - 0.50
assert round(profit, 2) == 0.70
```

The point estimate is positive while the contribution interval includes losses.

Power Begins With the Useful Effect

- Choose the smallest effect that could change the action.
- Use the expected outcome variance and assignment unit.
- Halving the target effect approximately quadruples sample size.

Implementation Checks

- Reconcile assigned counts and intended allocation.
- Check missing outcomes and tracking by assignment.
- Preserve the planned outcome horizon.
- Investigate exclusions that depend on treatment response.

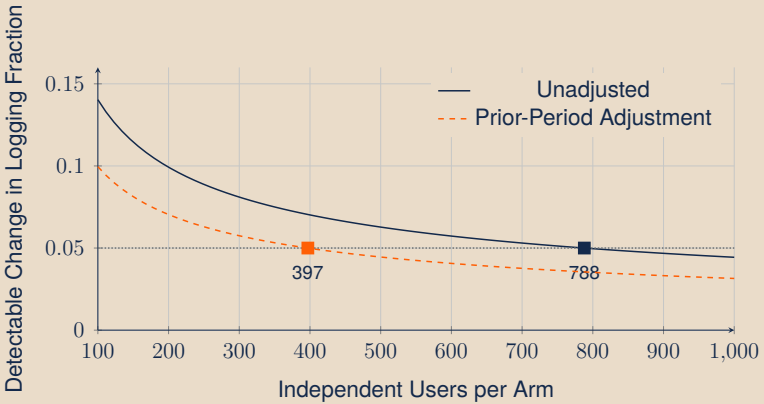
Prior Outcomes and Precision

Adjust the outcome using a pre-treatment measure X.

$$Y_i^{\text{adj}} = Y_i - \theta(X_i - \bar{X})$$

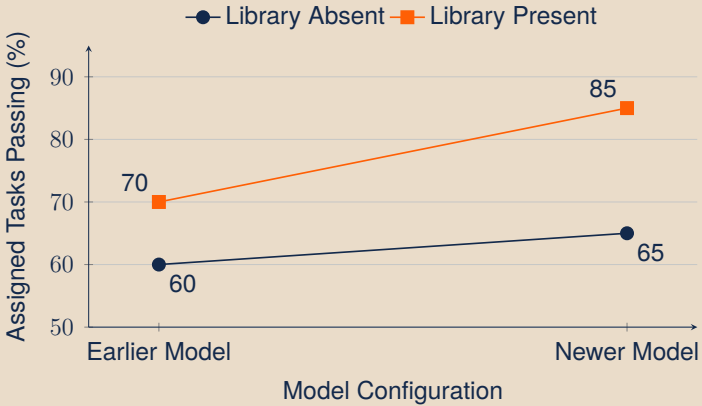
A predictive baseline can reduce residual variation while preserving the assignment comparison.

mHealth Baseline Activity Improves Precision



Planning calculation uses historical user-level variance and a five-point target.

Model and Library Contributions



Synthetic counts: 100 assigned tasks per condition; common rubric and resource limits.

An Interaction Calculation

The four rates are imposed teaching inputs.

```
q00, q01, q10, q11 = 0.60, 0.70, 0.65, 0.85
model_alone = q10 - q00
library_after_model = q11 - q10
interaction = (q11-q10) - (q01-q00)
assert round(interaction, 2) == 0.10
assert round(model_alone + library_after_model, 2) == 0.25
```

A joint gain does not uniquely allocate the interaction to either component.

Coding Exercise: An Experiment Plan

- Compute the factorial contrasts alongside the email effect.
- Change contribution, delivery cost, and the target effect.
- Specify an mHealth recommendation assignment with a fixed user denominator.

Interference and Assignment Units

- Customers can share offers or affect one another.
- Stores can divert purchases from nearby locations.
- The assignment unit should reflect the likely spillovers.

A Treatment Arm Needs a Defensible Design

- Random assignment identifies a comparison.
- The content of each treatment still affects the customers who receive it.
- Include customer outcomes, refusal, and stopping conditions.

Experiment Discussion

- Would a conversion gain cover the campaign cost?
- What changes if only email openers enter the analysis?
- Which guardrail outcome could reverse the rollout decision?